

Geometry-Aware Coherent Registerization of Single-Qubit Data Re-Uploading Models

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Single-qubit data re-uploading models are usually terminated by measuring a fixed observable. We study their use as coherent internal modules. The interface has two stages. First, a quaternion representation of $SU(2)$ motion is used to separate gauge-only motion from observable state motion, after which a gauge-aware geometric construction selects a signed Bloch coordinate. Second, the selected coordinate is mapped to a projector probability, estimated coherently, decoded into a signed fixed-point register, and used to control a downstream quantum evolution without intermediate measurement. We prove the quotient-distance identity, the optimality and identifiability conditions of the tangent-axis estimator, and the registerization and downstream error bounds. Numerical controls validate the gauge correction and projection blindness. A complete Qiskit demonstrator implements coherent amplitude estimation, reversible phase-to-sign decoding, register-controlled evolution, and uncomputation. The construction establishes composability, but low-precision QAE remains strongly input dependent and the finite truth-table decoder is not scalable.

1 Introduction

A single-qubit Quantum Re-uploading Unit (QRU) prepares

$$|\psi_\theta(x)\rangle = U_\theta(x)|0\rangle, \quad U_\theta(x) \in SU(2). \quad (1)$$

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Data re-uploading circuits are expressive one-qubit learners whose accessible frequency structure depends on their encoding generators and architecture [1, 2, 3]. They are nevertheless usually treated as terminal models: a fixed observable, commonly Z , is measured and converted into a classical scalar.

A composed quantum pipeline requires a different interface: the QRU output must be selected as a physically meaningful signed coordinate and transported coherently to a downstream routine. Figure 1 summarizes the resulting architecture. The work does not introduce a new amplitude-estimation algorithm and makes no quantum-advantage claim.

The first contribution is a gauge-aware signed-readout construction based on the observable QRU trajectory. The second is a coherent finite-precision interface that maps the selected readout to a quantum register and uses that register as a downstream control.

2 Background and related work

The QRU state has Bloch vector

$$\mathbf{r}_\theta(x) = (\langle X \rangle_x, \langle Y \rangle_x, \langle Z \rangle_x)^T \in \mathbb{S}^2. \quad (2)$$

For a unit axis $\mathbf{v} \in \mathbb{S}^2$, define the signed readout

$$s_{\mathbf{v}}(x) = \mathbf{v}^T \mathbf{r}_\theta(x) = \langle \psi_\theta(x) | \mathbf{v} \cdot \boldsymbol{\sigma} | \psi_\theta(x) \rangle \in [-1, 1]. \quad (3)$$

The conventional readout is $s_{e_z} = \langle Z \rangle$.

Every $U \in SU(2)$ admits the quaternion representation

$$U = \begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix}, \quad \alpha = q_0 - iq_3, \quad \beta = -q_2 - iq_1, \quad (4)$$

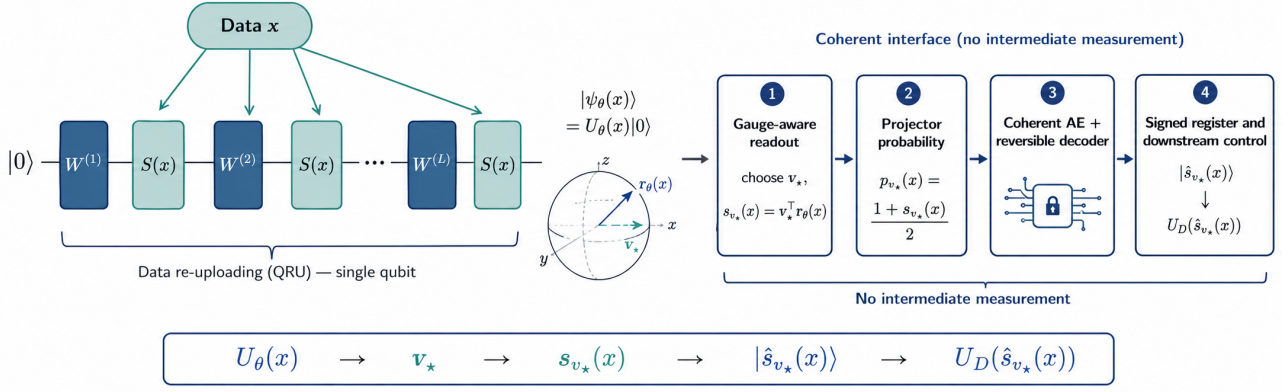


Figure 1: Overview of the proposed QRU interface. A single-qubit data re-uploading model prepares $|\psi_\theta(x)\rangle = U_\theta(x)|0\rangle$. A gauge-aware geometric readout selects the signed Bloch coordinate to expose, which is then converted into a projector probability, estimated coherently, decoded into a signed register and used as a downstream quantum control without intermediate measurement.

with $q \in \mathbb{S}^3$ and $q \sim -q$. The unitary acts on Bloch vectors through the associated rotation in $SO(3)$, but the prepared pure state belongs to

$$SU(2)/U(1) \simeq \mathbb{CP}^1 \simeq \mathbb{S}^2. \quad (5)$$

Consequently, motion of the full unitary need not be observable in $U|0\rangle$.

For a selected axis, the directional projector

$$\Pi_v^+ = \frac{\mathbb{I} + \mathbf{v} \cdot \boldsymbol{\sigma}}{2} \quad (6)$$

obeys

$$p_v(x) = \langle \psi_\theta(x) | \Pi_v^+ | \psi_\theta(x) \rangle = \frac{1 + s_v(x)}{2}. \quad (7)$$

Coherent amplitude estimation can encode information about p_v in a phase register using controlled oracle calls [4, 5, 6]. Standard phase-estimation-based QAE returns a word $y \in \{0, \dots, 2^{m_\phi} - 1\}$ whose amplitude estimate is

$$\hat{p}_y = \sin^2\left(\frac{\pi y}{2^{m_\phi}}\right). \quad (8)$$

This phase word is not a linear binary representation of p_v or s_v ; coherent use by another algorithm requires a reversible decoder.

3 Gauge-aware signed-readout selection

3.1 Quotient geometry

The relative rotation angle of two unit quaternions is

$$\delta_{\text{rot}}(q_i, q_j) = 2 \arccos(|q_i^T q_j|). \quad (9)$$

It is invariant under $q \mapsto -q$ but not under the stabilizer of $|0\rangle$. Define the quotient distance

$$\delta_{\text{quot}}(U_i, U_j) = \min_{\phi} \delta_{\text{rot}}(U_i, U_j R_z(\phi)). \quad (10)$$

Proposition 1 (Quotient-state identity). *For $|\psi_k\rangle = U_k|0\rangle$,*

$$\delta_{\text{quot}}(U_i, U_j) = 2 \arccos(|\langle \psi_i | \psi_j \rangle|) = \arccos(\mathbf{r}_i^T \mathbf{r}_j). \quad (11)$$

The first equality removes exactly the right $U(1)$ action that changes only the global phase of $U|0\rangle$. The second identifies this quotient with the pure-qubit Bloch geodesic distance. A proof is given in Appendix A.

The control $U(x) = V R_z(x)$ exposes the necessity of the quotient. Its raw rotation accumulates 2π , while its quotient and Bloch-geodesic distances vanish numerically; see Fig. 2(a).

3.2 Tangent-axis estimator

For an ordered grid $x_1 < \dots < x_N$, let

$$D_i = \frac{\mathbf{r}_{i+1} - \mathbf{r}_i}{\Delta x_i}, \quad w_i \geq 0, \quad (12)$$

and define

$$C_w = \frac{\sum_i w_i D_i D_i^T \Delta x_i}{\sum_i w_i \Delta x_i}. \quad (13)$$

The state-motion construction uses $w_i = \delta_{\text{quot}}(U_i, U_{i+1})^2$, which equals Bloch-geodesic weighting by Proposition 1. Raw quaternion weights are retained only as an ablation.

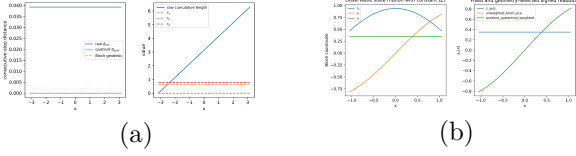


Figure 2: Geometric controls. (a) Gauge-only $SU(2)$ motion produces raw rotation length while all Bloch coordinates remain constant. (b) A constant Z readout hides observable motion in the XY plane.

Proposition 2 (Optimal axis). *For $\|\mathbf{v}\| = 1$, the energy*

$$E_w(\mathbf{v}) = \mathbf{v}^T C_w \mathbf{v} \quad (14)$$

is maximized by a principal eigenvector of C_w .

If $\lambda_1 \geq \lambda_2 \geq \lambda_3$ are the eigenvalues, we report

$$\Gamma = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \eta}. \quad (15)$$

The axis is rejected when λ_1 or Γ falls below a fixed tolerance. Proposition 2 follows directly from Rayleigh–Ritz; the rejection condition prevents an arbitrary eigenvector from being interpreted as a physical direction when the covariance is null or degenerate. Since a principal eigenvector is defined up to sign, the signed interface uses the canonical orientation $k = \arg \max_j |v_j|$ and $\mathbf{v} \leftarrow -\mathbf{v}$ if $v_k < 0$.

3.3 Validation and scope of the geometric claim

The second analytic control uses $U(x) = R_z(x)R_y(\theta)$ on a restricted arc. Its Z coordinate is constant while its state moves in the XY plane. The identifiable geometric estimators recover an XY direction, whereas Z has zero tangent energy; see Fig. 2(b).

For random QRUs, axes are fitted on $[-\pi, 0]$ and evaluated on $[0, \pi]$ using one common state-motion energy. Quotient weighting is state-motion weighting by Proposition 1. Relative to unweighted Bloch PCA, its paired mean test-energy gain has a bootstrap interval excluding zero only at depth two; intervals include zero at depths three, five, and eight (Fig. 4(a)). The quotient construction is therefore a gauge-correct derivation of the state geometry, not evidence of a general quaternion-specific performance advantage.

The Bloch-register diagnostic in Fig. 3 uses one deterministic depth-eight QRU instance selected

by the same identifiability test. For this instance,

$$\mathbf{v}_\star = (-0.5939, 0.6573, -0.4641)^T, \quad \Gamma = 0.3955. \quad (16)$$

and

$$\max_x \left| p_{\mathbf{v}_\star}(x) - \frac{1 + s_{\mathbf{v}_\star}(x)}{2} \right| = 2.5 \times 10^{-16}. \quad (17)$$

The figure is therefore a checked interface diagnostic, not a supervised-performance result.

4 Coherent signed-register interface

4.1 Signed registerization and error bounds

We use canonical signed magnitude with one sign bit and m_s fractional magnitude bits:

$$\hat{s} = (-1)^b \sum_{k=1}^{m_s} b_k 2^{-k}, \quad (18)$$

with sign bit zero for the numerical value zero and saturation at $1 - 2^{-m_s}$.

Proposition 3 (Registerization error). *Assume $|\tilde{p} - p| \leq \epsilon_p$, set $\tilde{s} = 2\tilde{p} - 1$, and quantize $\tilde{s}$ to $\hat{s}$. Then*

$$|\hat{s} - s| \leq 2\epsilon_p + 2^{-m_s}. \quad (19)$$

For $U_D(s) = e^{-i\gamma s Z}$,

$$\|U_D(s) - U_D(\hat{s})\| = 2 \left| \sin \frac{\gamma(s - \hat{s})}{2} \right| \leq |\gamma| |s - \hat{s}|. \quad (20)$$

The first inequality combines affine error propagation with fixed-point quantization; the second follows from the spectrum of Z and $|\sin u| \leq |u|$.

4.2 Coherent construction

Choose B_v such that

$$B_v^\dagger Z B_v = \mathbf{v} \cdot \boldsymbol{\sigma}. \quad (21)$$

For $\mathbf{v} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, one valid choice is $B_v = R_y(-\theta)R_z(-\phi)$. A standard QPE-based QAE writes a coherent phase word y . A reversible finite decoder implements

$$\begin{aligned} |y\rangle|0\rangle &\mapsto |y\rangle|\hat{s}_y\rangle, \\ \hat{s}_y &= Q_{m_s} \left[2 \sin^2 \left(\frac{\pi y}{2^{m_\phi}} \right) - 1 \right]. \end{aligned} \quad (22)$$

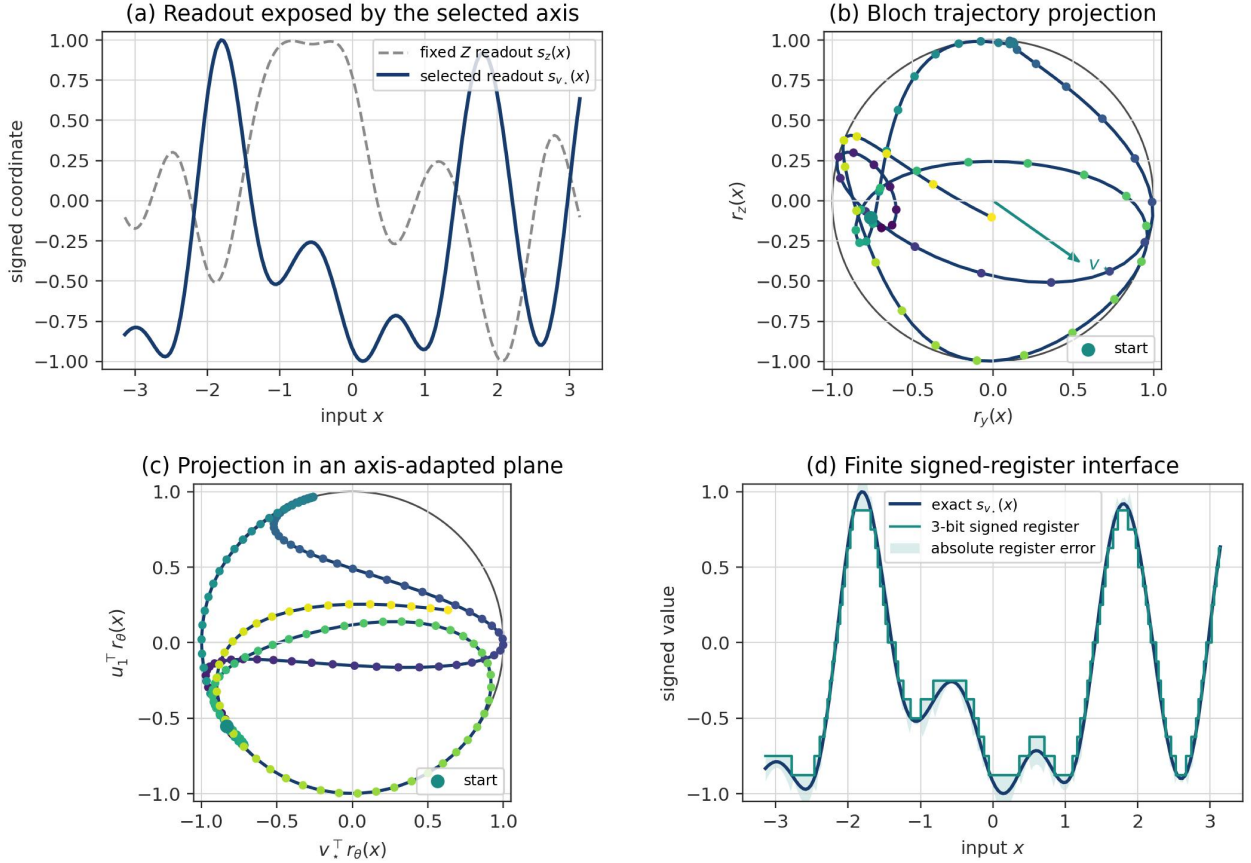


Figure 3: Bloch-register diagnostic. The selected axis has $2.35\times$ the tangent energy of e_z ; the three-bit signed register has mean absolute error 5.88×10^{-2} .

The signed register controls

$$U_D^{\text{reg}} = \sum_z |z\rangle\langle z| \otimes e^{-i\gamma s_z Z}. \quad (23)$$

Because all Z terms commute, each magnitude bit controls one weighted rotation and the sign bit reverses its direction. The signed register is then uncomputed. Circuit details are summarized in Appendix B.

4.3 Functional validation, precision, and resources

All signed words for $m_s = 2, 3$ reproduce the direct rotation with state infidelity below 4×10^{-15} . A seven-qubit two-input construction maintains a superposition of QRU branches, writes two signed values coherently, applies the downstream evolution, and reverses the pipeline. Its interference echo gives $P(0) = 1$ to numerical precision, while interface dephasing gives $P(0) = 1/2$.

The complete QAE-decoder-downstream circuit is validated against a direct phase-word-controlled reference. For $(m_\phi, m_s) = (4, 3)$, it uses ten qubits and attains fidelity above $1 - 1.4 \times 10^{-13}$ with signed-register reset probability above $1 - 1.3 \times 10^{-13}$. The resources in Table 1 are regenerated from the circuit. The finite lookup, together with its uncomputation, is the dominant cost.

The favourable point $p = 0.3059$ is not representative. Over the uniform probability grid, only 1.5%, 4.5%, and 11.4% of values achieve signed error at most 0.02 for $(3, 2)$, $(4, 3)$, $(5, 4)$. The oscillatory phase-grid error and its downstream effect are shown in Fig. 4(b).

5 Discussion

The two contributions solve distinct parts of one composability problem. Geometry determines which scalar coordinate is exposed; coherent reg-

Table 1: Coherent-interface precision and resources. Transpilation uses basis gates $\{\text{rz}, \text{sx}, \text{x}, \text{cx}\}$, optimization level one, and seed 42. Decoder CX counts one phase-to-signed lookup; the inverse lookup has comparable cost.

(m_ϕ, m_s)	qubits	depth	CX	decoder CX	grid mean $ \hat{s} - s $	QRU mean $ \hat{s} - s $
(3, 2)	8	795	326	150	0.244	0.242
(4, 3)	10	6156	2170	1050	0.139	0.139
(5, 4)	12	19443	6513	3210	0.0829	0.0816

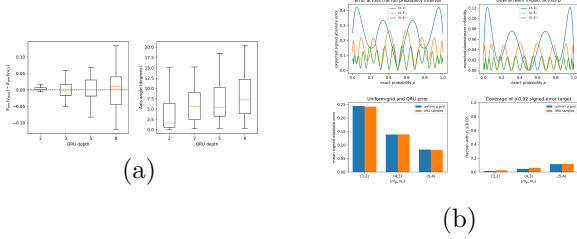


Figure 4: Main numerical results. (a) Out-of-sample quotient/state weighting versus unweighted Bloch PCA. (b) Finite-precision QAE and signed-decoder error over the probability domain and QRU samples.

isterization transports that coordinate without an intermediate classical interface. The geometric study also limits its own claim: quotient quaternion motion equals Bloch-geodesic state motion, and current random-QRU experiments do not establish systematic improvement over Bloch PCA. The remaining value of the quaternion formulation is the explicit diagnosis and removal of gauge-only motion.

The coherent circuit establishes functional composability, not near-term practicality. A measurement-reinjection pipeline may be preferable for one known classical input. Coherent transport becomes structurally relevant when inputs, scenarios, or downstream branches remain in superposition, as in the two-branch interference test. Even then, standard low-precision QAE has strongly input-dependent errors, while the exact finite decoder scales as a lookup over 2^{m_ϕ} phase words. A scalable implementation requires a structured reversible approximation of the sine-square map or an estimator producing a more directly consumable numerical representation. The present results therefore support an interface theorem and a verified small demonstrator, not a scalable NISQ architecture.

6 Conclusion

A QRU can be used as an internal quantum module only after defining both a meaningful signed output and a coherent transport mechanism. We introduced a gauge-correct tangent-axis construction, proved its quotient-state interpretation and spectral optimality, and showed that it detects projection blindness without supporting a general quaternion-specific advantage over Bloch PCA. We then implemented a coherent QRU-to-phase-to-signed-register-to-downstream pipeline with reversible decoding and uncomputation. The circuit is functionally correct and preserves interference, but its low-precision error and truth-table resource growth delimit the current claim. The next technical requirement is a structured reversible decoder and validation on trained QRUs.

Acknowledgment and Artifact Availability

This work was conducted within the framework of the TAI AO project and University of Waikato. A reference implementation is available at: <https://github.com/LeaCasse/qru-quaternion-registerization>.

A Proofs

Proof of Proposition 1. For unit quaternions $q_i, q_j \in \mathbb{S}^3$, let $q_{ij} = q_i^{-1}q_j$. Since $q_i^{-1} = \bar{q}_i$ and both quaternions are unit,

$$\text{Re}(q_{ij}) = q_i^T q_j. \quad (24)$$

Writing the relative rotation as $q_{ij} = \cos(\theta/2) + \mathbf{n} \sin(\theta/2)$ gives $\theta = 2 \arccos |q_i^T q_j|$. The absolute value implements the projective identification $q \sim -q$. Hence δ_{rot} measures the relative rotation in $SU(2)/\{\pm I\} \simeq SO(3)$; it is not invariant under the right stabilizer of $|0\rangle$.

Let $W = U_i^\dagger U_j$. Right multiplication by $R_z(\phi)$ changes the prepared state by

$$U_j R_z(\phi) |0\rangle = e^{-i\phi/2} U_j |0\rangle, \quad (25)$$

so minimizing over ϕ removes exactly the vertical $U(1)$ fibre of the ray $[U_j|0\rangle]$. The induced quotient distance between the two prepared rays is therefore

$$\delta_{\text{quot}}(U_i, U_j) = 2 \arccos |\langle \psi_i | \psi_j \rangle|. \quad (26)$$

For pure qubits,

$$|\langle \psi_i | \psi_j \rangle|^2 = \frac{1 + \mathbf{r}_i^T \mathbf{r}_j}{2}. \quad (27)$$

The half-angle identity then gives

$$2 \arccos |\langle \psi_i | \psi_j \rangle| = \arccos(\mathbf{r}_i^T \mathbf{r}_j), \quad (28)$$

which is the Bloch geodesic distance. $\square$

Proof of Proposition 2. In the continuous setting, the weighted readout-variation energy is

$$E_w(\mathbf{v}) = \frac{\int w(x) \left| \frac{d}{dx} s_{\mathbf{v}}(x) \right|^2 dx}{\int w(x) dx}, \quad \|\mathbf{v}\| = 1. \quad (29)$$

Since $s_{\mathbf{v}}(x) = \mathbf{v}^T \mathbf{r}_{\theta}(x)$,

$$\frac{d}{dx} s_{\mathbf{v}}(x) = \mathbf{v}^T \mathbf{r}'_{\theta}(x), \quad E_w(\mathbf{v}) = \mathbf{v}^T C_w \mathbf{v}, \quad (30)$$

with

$$C_w = \frac{\int w(x) \mathbf{r}'_{\theta}(x) \mathbf{r}'_{\theta}(x)^T dx}{\int w(x) dx}. \quad (31)$$

The discrete estimator in Eq. (14) is its finite-difference quadrature form. Since C_w is symmetric positive semidefinite, the Rayleigh–Ritz theorem yields

$$\max_{\|\mathbf{v}\|=1} \mathbf{v}^T C_w \mathbf{v} = \lambda_1, \quad (32)$$

attained by any principal eigenvector. If $\lambda_1 = 0$ the trajectory has no observable tangent energy; if $\lambda_1 = \lambda_2$ the maximizing axis is not unique. The reported eigengap Γ enforces identifiability before the axis is used as a signed interface. $\square$

Proof of Proposition 3. Since $s = 2p - 1$ and $\tilde{s} = 2\tilde{p} - 1$, $|\tilde{s} - s| = 2|\tilde{p} - p| \leq 2\epsilon_p$. Signed-magnitude truncation with m_s fractional bits has error at most 2^{-m_s} , including endpoint saturation. The triangle inequality yields the first result. Diagonalizing Z gives $\|e^{-i\gamma s Z} - e^{-i\gamma \hat{s} Z}\| = \max_{z=\pm 1} |e^{-i\gamma s z} - e^{-i\gamma \hat{s} z}| = 2|\sin(\gamma(s - \hat{s})/2)|$, which is bounded by $|\gamma||s - \hat{s}|$. $\square$

B Coherent circuit construction

Let $A_{\mathbf{v}} = X B_{\mathbf{v}} U_{\theta}(x)$ so that $|\langle 1 | A_{\mathbf{v}} | 0 \rangle|^2 = p_{\mathbf{v}}(x)$. The Grover operator is

$$Q_{\mathbf{v}} = -A_{\mathbf{v}} S_0 A_{\mathbf{v}}^\dagger S_{\text{good}}, \quad (33)$$

where $S_0 = \mathbb{I} - 2|0\rangle\langle 0|$ and $S_{\text{good}} = \mathbb{I} - 2|1\rangle\langle 1|$. QPE applied to $Q_{\mathbf{v}}$ yields the symmetric phases associated with $p_{\mathbf{v}} = \sin^2 \theta$. The finite decoder is a reversible lookup over the 2^{m_ϕ} phase words. It is exact for the selected discretization but has exponential synthesis cost. After the controlled downstream operation, applying the decoder again returns all signed ancillas to $|0\rangle$.

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